FGE RENDER REALISM PLAYBOOK

Grok Generation — Master Instruction

Agent Skill Module: RENDER-001

> *"The difference between a good image and a real one is knowing which renderer built it in your mind."*

WHAT THIS PLAYBOOK DOES

This playbook teaches Grok to think like a professional renderer — not just generate images, but construct them with the same physical logic that V-Ray, Arnold, and OctaneRender use.

The result: FGE output that doesn't look AI-generated. It looks **produced.**

THE RENDERER DICTIONARY

Before generating anything, tell Grok which renderer to think like. Each one produces a different quality of truth.

V-RAY MODE

Use for: Hero shots, final card images, maximum realism

```
Invoke phrase:
"Think like V-Ray. Adaptive ray-tracing.
Calculate every light bounce physically.
Material properties are non-negotiable.
This is a hero shot."

What it does:
- Every light ray traced to source
```

```
- Material SSS calculated not approximated
- Shadow quality: physically accurate
- Gold reflects gold. Chrome reflects environment.
- Nothing faked. Nothing baked.

Best for:
- Zen Nowhere final portrait
- Raven Voss card hero image
- Talisman object render
- Any card that is the definitive edition
```

ARNOLD MODE

Use for: Cinematic character shots, Hollywood quality

```
Invoke phrase:
"Think like Autodesk Arnold.
Cinematic pipeline quality.
Handle scene complexity without compromise.
Predictable, visually stunning, film-grade."

What it does:
- Depth of field as a narrative tool
- Skin SSS at film quality
- Volumetric atmosphere (dust, haze, smoke)
- Character lighting like a cinematographer
- aiStandardSurface quality on every material

Best for:
- Zen Nowhere cinematic portrait
- Scene atmosphere renders
- Any character in environmental context
- The moment before the Scene Law activates
```

OCTANE MODE

Use for: GPU-accelerated spectral accuracy, product renders

```
Invoke phrase:
"Think like OctaneRender.
Spectral rendering — light behaves as physics.
GPU-accelerated unbiased render.
```

```
Every wavelength calculated."

What it does:
- Spectral light calculation (real wavelengths)
- Gemstone and crystal renders at maximum accuracy
- Chrome and metal: physically perfect
- Opal, labradorite, fire opal: full spectral play
- Subsurface on pearl and marble: volumetric

Best for:
- Talisman gem render
- All gemstone materials
- Liquid chrome on Raven Voss
- Any material where light physics matter
```

CORONA MODE

Use for: Environment, interior, atmospheric scenes

```
Invoke phrase:
"Think like Chaos Corona.
Artist-friendly physically based lighting.
Interior visualization quality.
Lighting realism, minimal technical noise."

What it does:
- Interior light behavior: bounces, fills, bleeds
- Atmosphere: the room has air in it
- Material warmth: everything feels touchable
- Lighting tells the story

Best for:
- The room Zen Nowhere walks into
- Raven Voss penthouse scene
- Any environment card background
- The scene before Status Collapse activates
```

UNREAL ENGINE 5 MODE

Use for: World-building, navigable scenes, real-time

```
Invoke phrase:
```

```
"Think like Unreal Engine 5 with Lumen.
Dynamic global illumination.
Nanite geometric detail.
Real-time world quality."

What it does:
- Lumen: light responds to scene in real time
- Nanite: geometric complexity without loss
- World feels navigable, not staged
- Everything exists in space, not on a surface

Best for:
- FGE world environment renders
- Scene-setting card backgrounds
- Vegas world establishing shots
- Any render that needs to feel like a place
```

GAUSSIAN SPLAT MODE

Use for: Photo-to-3D, real-world scene capture

```
Invoke phrase:
"Think like Luma AI Gaussian Splatting.
Real-world photo converted to 3D splats.
Complex lighting and reflections captured.
The world captured not constructed."

What it does:
- Real-world photographic quality
- Lighting captured from actual environment
- Reflections physically present not simulated
- The scene feels found not made

Best for:
- Reference scene backgrounds
- Real-location inspired environments
- When the scene needs documentary truth
- Street scenes for Zen Nowhere's origin
```

THE REALISM STACK

For maximum output quality, layer these instructions in order. Every render gets all five

layers.

```
LAYER 1 — RENDERER MODE
"Think like [V-Ray / Arnold / Octane /
Corona / UE5 / Gaussian Splat]"

LAYER 2 — PHYSICAL LIGHT
"Light source: [position + quality]
Light behaves physically —
bounces, fills shadows, bleeds color.
No flat lighting. No baked shadows."

LAYER 3 — MATERIAL PHYSICS
"[Material name] — physically accurate:
[SSS level] + [reflection type] +
[surface texture] + [displacement depth]
Material is not a texture applied to a shape.
It is a physical substance with depth."

LAYER 4 — ATMOSPHERIC TRUTH
"The air in this scene has:
[dust / humidity / heat haze / rain /
night atmosphere / golden hour particles]
Atmosphere is volumetric.
It affects everything in the scene."

LAYER 5 — FGE CONSTITUTIONAL LAYER
"This exists inside Feral Gloss Empire.
Aesthetic law: rugged but rich.
Every surface has survived something.
The tension between opposites is the render."
```

THE QUICK GUIDE — RENDERER BY GOAL

Direct translation of the research into FGE deployment decisions:

```
GOAL                    RENDERER MODE

Maximum card realism    V-Ray
Hero character shot     V-Ray or Arnold
Cinematic atmosphere    Arnold
Gemstone accuracy       Octane
Chrome/metal quality    Octane
Interior scene          Corona
```

```
World environment        UE5 Lumen
Street reality           Gaussian Splat
Fast iteration           D5 Render logic
Product close-up         OctaneRender
```

PROMPT FORMULAS BY USE CASE

FORMULA 1 — CHARACTER HERO SHOT

For final card-grade character renders

```
"Think like Arnold cinematic pipeline.
Film-grade character lighting.

Subject: [character name + full description]

Renderer instructions:
- Skin SSS: film quality depth
- Single key light: [position]
- Fill: subtle, lifted shadows
- Rim: defines silhouette from background
- Atmosphere: [dust / haze / night air]
- Depth of field: shallow, subject sharp,
  environment suggests

Material physics:
- [clothing material] behaves as [physics]
- [skin] subsurface scattering: warm depth
- [accessories] specular: [quality]

FGE layer:
Rugged but rich. Every surface earned.
Cinematic truth. Not posed — present.

Photorealistic. 8K. Film grain optional."
```

FORMULA 2 — MATERIAL CLOSE-UP

For texture and material hero renders

```
"Think like OctaneRender.
```

```
Spectral light calculation.
GPU-accelerated physical accuracy.

Material: [full material specification]

Render instructions:
- Light source: single directional
  from [angle] degrees
- Surface response: physically calculated
- SSS: [level] — light enters and exits
- Reflection: [mirror / diffuse / hybrid]
- Displacement: micro surface irregularity
- Highlight: [sharp / soft / none]

Camera: macro distance, shallow DOF
Surface fills frame.
One edge catches maximum light.
Opposite edge falls to shadow.

FGE: Material has history.
Not a sample. A surface that survived."
```

FORMULA 3 — SCENE ENVIRONMENT

For world-building and background cards

```
"Think like Unreal Engine 5 with Lumen.
Dynamic global illumination.
The world exists. We are inside it.

Scene: [environment description]

Lumen instructions:
- Light source: [sun position / artificial]
- Global illumination: bounces naturally
- Sky: [time of day + atmosphere]
- Ground: [material + condition]
- Distance: atmospheric perspective present

Nanite detail:
- Geometric complexity maintained
- Distance objects retain detail
- No LOD visible

FGE world layer:
```

```
This is where [character name] exists.
The world is real before they arrive.
Their presence changes it."
```

FORMULA 4 — CARD COMPOSITE

For full trading card renders

```
"Think like V-Ray hero shot.
Maximum physical accuracy.
This is the definitive edition.

Card composition:
- Character: [full specification]
- Background: [material + environment]
- Border: gold foil frame, embossed
- Lighting: [key + fill + atmosphere]
- Edition marker: [holographic / foil / standard]

Layer physics:
- Character exists IN the environment
  not placed ON a background
- Light source is consistent across
  character + background + border
- Atmosphere connects all elements

Card text:
- Name plate: [character name]
- Stats block: [relevant stats]
- Edition: [ET-number — year — #/100]
- Trade line: FERAL GLOSS EMPIRE

FGE: This card is a window into
the world. Not an illustration of it."
```

FORMULA 5 — ATMOSPHERIC SCENE

For mood and environment storytelling

```
"Think like Chaos Corona.
Artist-friendly physically based atmosphere.
The room has air. The scene breathes.
```

```
Scene: [location + time + mood]

Atmospheric instructions:
- Air density: [clear / dusty / humid /
  smoky / rainy / night city]
- Light source: [position + quality + color]
- Bounce light: fills shadows naturally
- Color bleed: materials affect nearby surfaces
- Depth fog: atmosphere has distance

Storytelling layer:
Something is about to happen.
The atmosphere knows before the characters do.
Anticipatory still.

FGE: [character name] is either
present or about to arrive.
The scene has already changed."
```

THE REALISM CHECKLIST

Before finalizing any Grok generation, run this check. Every item is a yes or no.

```
LIGHT
☐ Is there a defined light source position?
☐ Does light behave — bounce, fill, bleed?
☐ Are shadows physically placed not painted?
☐ Does the atmosphere have volumetric quality?

MATERIAL
☐ Is every material named precisely?
☐ Does SSS apply where it should?
☐ Are reflections environment-mapped?
☐ Does surface displacement exist?
☐ Is there micro-texture at close range?

CHARACTER
☐ Does the character exist IN the scene?
☐ Is the light source consistent on them?
☐ Do their materials respond to the environment?
☐ Is their expression constitutionally correct?

ATMOSPHERE
```

```
☐ Does the air have quality?
☐ Is depth visible — near vs far?
☐ Does the atmosphere serve the narrative?
☐ Is the anticipatory quality present?

FGE CONSTITUTIONAL
☐ Does it carry tension between opposites?
☐ Has every surface survived something?
☐ Would this offend someone?
☐ Would it obsess someone else?
☐ Does it feel like the world — not AI?
```

EMERGING TOOLS — FGE PIPELINE NOTES

From the research — these are next-level additions for when the system expands:

```
CHAOS VANTAGE
→ Real-time V-Ray scene exploration
→ No light baking required
→ Future use: real-time card scene preview
→ FGE application: walk through the world
  before committing to final render

NVIDIA OMNIVERSE
→ Collaborative multi-tool 3D hub
→ Future use: multi-agent scene building
→ FGE application: multiple agents
  building the same world simultaneously

LUMA AI / KIRI ENGINE
→ Photo to Gaussian Splat 3D
→ Future use: real location capture
→ FGE application: real Vegas locations
  become navigable FGE world environments

D5 RENDER
→ Real-time ray tracing, fast iteration
→ Built-in AI tools
→ FGE application: rapid card background
  iteration before final V-Ray hero render
```

THE GROK SESSION PROTOCOL

Every render session follows this structure:

```
OPEN:
"You are the visual render engine
for Feral Gloss Empire.
Renderer mode: [chosen mode]
Constitutional law active.
Session goal: [specific output]"

GENERATE:
Use the appropriate formula above.
Layer all five realism layers.

EVALUATE:
Run the realism checklist.
If any box is unchecked — refine.

REFINE LANGUAGE:
Too flat →
"Add volumetric atmosphere.
The air in this scene has weight."

Too clean →
"Age it. Show what survived.
Every surface has history."

Too digital →
"Think [renderer name].
This is physical not procedural."

Wrong light →
"Single source from [position].
Every other light is a bounce
from that source. Nothing added."

Off-brand →
"Feral Gloss Empire.
Rugged but rich.
The tension IS the render."

CLOSE:
Save winning prompt.
Document: material + renderer + what worked.
Add to FGE material library.
```

RENDERER MODE QUICK CARDS

Copy these single lines into any Grok session to instantly shift render quality:

```
V-RAY:
"Physical accuracy non-negotiable.
Adaptive ray-tracing. Hero shot quality."

ARNOLD:
"Cinematic pipeline. Hollywood quality.
Scene complexity handled. Film-grade."

OCTANE:
"Spectral rendering. GPU-accelerated.
Every wavelength physically calculated."

CORONA:
"Artist-friendly physically based.
Interior atmosphere. Lighting realism."

UE5 LUMEN:
"Dynamic global illumination.
Nanite geometry. The world is real."

GAUSSIAN SPLAT:
"World captured not constructed.
Photographic truth. Splat quality."
```

MODULE IDENTITY

```
Module:     RENDER-001
Type:       Skill Module —
            Render Realism Playbook
Universe:   Feral Gloss Empire
Authority:  FGE Constitutional Law
            ET-CON-000001
Engine:     Grok Primary
Version:    1.0 — 2026
Status:     Active
```

```
Invoke:     "Render Agent, active.
             Mode: [renderer]"

Combine with:
→ TEX-AGENT-001 for material sourcing
→ FGE Constitution for character law
→ Material Universe for prompt blocks
```

THE SINGLE MOST IMPORTANT INSIGHT

```
Every renderer on this list
does the same thing differently:

It asks:
"What would light actually do here?"

That is the question to ask Grok
before every generation.

Not: "Make this look good."
But: "What would light actually do
      to this surface, in this atmosphere,
      at this moment?"

When Grok answers that question
with physical accuracy —

the output stops looking generated.

It starts looking true.
```

RENDER-001 — FERAL GLOSS EMPIRE — 2026 PHYSICAL ACCURACY NON-NEGOTIABLE NO REFUNDS — FINAL SALE

"The render isn't finished when everything is added. It's finished when nothing is wrong."